

# pyRTX: a Python package high precision computation of non gravitational forces on deep space probes

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## Summary

With the constant improvement of radiometric tracking systems, inaccuracies in the non-gravitational force modeling have become one of the limiting factors to precise orbit determination, and the scientific products that it enables. The main factor impacting the limited accuracy of non-gravitational force models is the complex 3D shape of the spacecraft. While fast, reliable, analytical models are available for simple shapes (spheres, cubes, etc), no such model is generally available for a complex shape. This software package aims to address this limitation by leveraging ray-tracing to compute the complex interaction between the forcing environment (radiation, atmosphere) and the three dimensional shape of the spacecraft.

## Statement of need

Several scientific investigations require high-precision reconstruction of spacecraft trajectories. Among these, one of the most demanding is the determination of the gravity field of Solar System bodies (planets, moons). This task is accomplished by solving the so-called orbit determination (OD) problem (Milani & Gronchi, 2009; Tapley et al., 2004). The solution of the OD in the adjustment of a dynamical model (a set of differential equations) describing the spacecraft motion. Systematic errors in the dynamical model will almost inevitably lead to systematic errors in the solution. In the recent years significant improvements in radiometric tracking system, have led to more and more precise measurements of the spacecraft position and velocity (the input to the OD), thus requiring increasingly more accurate dynamical models (Asmar et al., 2005; Cappuccio et al., 2020; Mazarico et al., 2023). One of the major limitations of current dynamical modelling of deep-space probes consists in the complex interaction between the spacecraft shape and the atmosphere, and with radiative forces (solar radiation pressure, albedo, thermal infrared radiation). We developed the pyRTX software package to address this limitation. Leveraging the ray-tracing technique, originally developed in computer graphics, instead of relying on simplified macro-models (flat plates, cylinders, etc) pyRTX is able to use the actual 3D shape of the spacecraft (provided as, for example, an .obj file), to compute several non-gravitational accelerations.

pyRTX addresses the gap in open source solutions for a comprehensive modelling of several, important, non-gravitational forces. Being built around the SPICE astrodynamics library (Annex et al., 2020), pyRTX is intended to be a plug-in tool that can be used in existing OD codes, and applications.

## 37 Functionality

38 In this section we describe the main functionalities of the pyRTX software. All of these  
39 functionalities are discussed in the example library included in the code distribution. Our  
40 companion paper (Zurria et al., 2026) discusses an actual application of the pyRTX library for  
41 ameliorating the OD of NASA's Lunar Reconnaissance Orbiter.

- 42 ■ *Solar Radiation Pressure Modeling*: pyRTX computes the acceleration due to solar  
43 photons by casting rays from a pixel plane representing the incoming solar flux. The  
44 engine inherently accounts for self-shadowing (where spacecraft components block light  
45 from reaching others) and multiple reflections. Users can specify optical properties for  
46 each mesh face, allowing the software to simulate both specular and diffuse (Lambertian)  
47 reflections.
- 48 ■ *Planetary Radiation Pressure*: The software models the effects of radiation reflected  
49 (albedo) and emitted (thermal infrared) by planetary bodies. This includes the ability to  
50 use complex planetary shape models (e.g., topography from digital elevation models,  
51 DEMs) and spatially variable maps for albedo, emissivity, and surface temperature.
- 52 ■ *Eclipse and Shadow Function Analysis*: pyRTX can compute precise shadow functions  
53 during eclipse transitions. It supports advanced modeling features such as solar limb  
54 darkening, where the variation in intensity across the solar disk is integrated into the  
55 flux calculation.
- 56 ■ *Atmospheric Drag*: For low-altitude missions, the software calculates the effective  
57 aerodynamic cross-section of the spacecraft, providing inputs for atmospheric drag  
58 modeling.
- 59 ■ *Lookup Table (LUT) Generation*: To enable efficient integration with OD software,  
60 pyRTX can pre-compute accelerations over a grid of incident directions. This feature  
61 supports spacecraft with articulating components (e.g., solar arrays), allowing users  
62 to generate comprehensive LUTs that map acceleration vectors to specific spacecraft  
63 orientations and strongly reduce computation time.

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